

# A further conjecture on OEIS A243516, settled by the proof already in hand

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## Abstract

OEIS A243516 carries more than one conjecture. One of them has a proof; this paper settles a DIFFERENT one, and it needs no new model and no new engine – only the linear recurrence already established for the sequence. A claim about a C-finite sequence whose own characteristic polynomial is a multiple of the established one is decided by polynomial divisibility, and where the claim is an identity rather than a recurrence, by divisibility together with a finite window of agreement.

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## 1 The premise

The entry's first terms are 3, 5, 10, 25, 77, 280, 1156, 5267,  $\dots$ , with offset 1. The sequence satisfies  $a(n) = 17a(n-1) - 111a(n-2) + 355a(n-3) - 584a(n-4) + 468a(n-5) - 144a(n-6)$ , a linear recurrence of order 6 with characteristic polynomial

$$q(x) = x^6 - 17x^5 + 111x^4 - 355x^3 + 584x^2 - 468x + 144.$$

This recurrence is proved in this project; it is the premise of everything below and nothing else is assumed. It was confirmed against every term the entry publishes from the index where it begins to hold, with at least 6 + 2 further terms after that index.

## 2 The further conjecture

Separately from the conjecture already settled, the entry records

$$\frac{a(n) = (2722 + 675 \cdot 2^n + 25 \cdot 2^{1+2n}) + 50 \cdot 3^{1+n} + 2^n \cdot 3^{1+n} + 660 \cdot n}{1800}.$$

The further claim is a CLOSED FORM, and for that the divisibility is not the whole argument. It gives that  $a$  and the closed form  $f$  satisfy the same linear recurrence; it does not say they are the same solution of it. Write  $p$  for the annihilator of  $f$ ,

$$p(x) = x^6 - 17x^5 + 111x^4 - 355x^3 + 584x^2 - 468x + 144,$$

of order  $D = 6$ . Since  $q \mid p$  and  $a$  satisfies the recurrence of  $q$ ,  $a$  satisfies that of  $p$  as well;  $f$  satisfies it by construction. Two solutions of a linear recurrence of order  $D$  with nonzero leading coefficient agree everywhere as soon as they agree at  $D$  consecutive indices, and  $a$  and  $f$  were checked to agree at  $n = 7, \dots, 12$ , which lies inside the terms the entry publishes and past the index from which  $a$  provably satisfies  $p$ . Hence  $a(n) = f(n)$  for every  $n$  in the claimed range.

**Theorem 1.** *The conjecture displayed above is true.*

### 3 What is and is not claimed here

This is a second result on an entry that already has one, and it is a second result only because it is a DIFFERENT statement. A generating function whose denominator's reciprocal is exactly  $q$  says precisely what the proved recurrence says, in another notation; such restatements are identified and are not counted. The conjecture settled here is not of that kind.